

22) Show that $p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent.

Soln: Let $p \rightarrow q$ be False. By defn. of conditional, p is True and q is False.

Let $\neg q \rightarrow \neg p$ be False. By defn. of conditional, $\neg q$ is True and $\neg p$ is False.

By defn. of negation, $\neg(\neg q)$ is False and $\neg(\neg p)$ is True.

By double-negation law, $\neg(\neg q) \equiv q$ and $\neg(\neg p) \equiv p$. $\therefore q$ is False and p is True.

$\therefore p \rightarrow q$ and $\neg q \rightarrow \neg p$ are logically equivalent. $\square$

23) Show that $\neg(p) \leftrightarrow q$ and $p \leftrightarrow \neg q$ are logically equivalent.

Soln: Let $\neg p \leftrightarrow q$ be True. By defn. of biconditional, either both $\neg p, q$ are True or both $\neg p, q$ are False. ~~or~~ If $\neg p$ is True, by defn. of negation $\neg(\neg p)$ is False. By double negation law, $\neg(\neg p) \equiv p$ is True. $\therefore p$ is False. Similarly we can show that if $\neg p$ is False, then p is True.

$\therefore \neg p \leftrightarrow q$ is True only if ① p is False, q is True OR ② p is True, q is False.

Let $p \leftrightarrow \neg q$ be True. By defn. of biconditional, either both $p, \neg q$ are True or both $p, \neg q$ are False. If $\neg q$ is True, by defn. of negation $\neg(\neg q)$ is False. By double negation law, $\neg(\neg q) \equiv q$ is True. $\therefore q$ is False. Similarly we can show that if $\neg q$ is False, then q is True.

$\therefore p \leftrightarrow \neg q$ is True only if ① p is True, q is False OR ② p is False, q is True. $\square$

24) Show that $\neg(p \oplus q)$ and $p \leftrightarrow q$ are logically equivalent.

Soln: Let $\neg(p \oplus q)$ be True. $\therefore$ By defn. of negation, $\neg(\neg(p \oplus q))$ is False. From double negation law, we know $\neg(\neg(p \oplus q)) \equiv p \oplus q$. $\therefore (p \oplus q)$ is False.

By defn. of XOR, $\therefore$ either both (p, q) are True or both (p, q) are False. Let $p \leftrightarrow q$ be True. $\therefore$ By defn. of biconditional, either both (p, q) are True or both (p, q) are False.

$\therefore \neg(p \oplus q)$ and $p \leftrightarrow q$ are logically equivalent $\square$